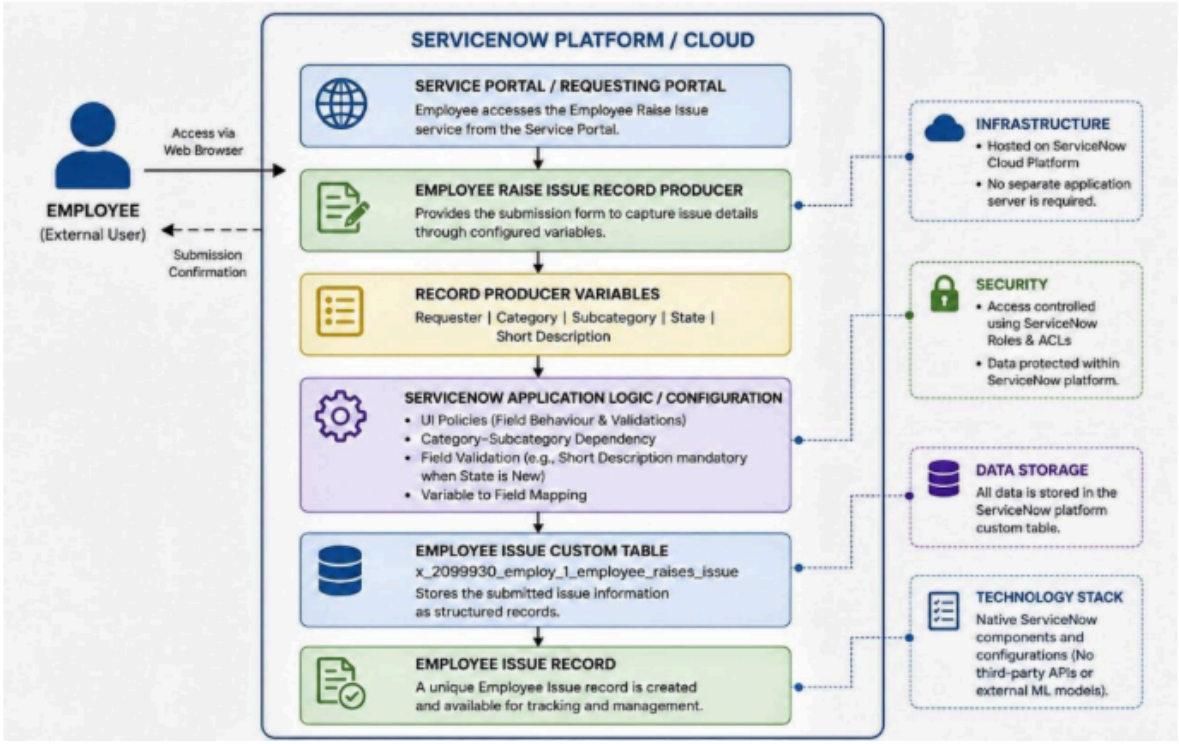
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	16 August 2026
Project Name	Employee Raise Issue – Record Producer & Service Portal Integration
Maximum Marks	4 Marks

Technical Architecture Overview

The "Employee Raise Issue" application is designed and deployed as a scoped, native solution within the ServiceNow platform. The architecture leverages a native, decoupled multi-layer framework—comprising the User Interface (UI) Layer, Application Logic Layer, and Data Services Layer—to ensure optimal performance, security, and scalability.



Here is the polished and professionally rewritten version of **Table-1**, **Table-2**, and the **References** section. The language has been upgraded to match enterprise-level technical

architecture and technical writing standards, utilizing precise ServiceNow framework terms (e.g., *User Criteria, Dictionary Overrides, Platform Native Engine*).

Table-1: Architecture Components & Technology Stack

S.No	Component	Functional Description	Core Technology / Platform Feature
1	User Interface (UI)	Employee-facing catalog portal utilized to search, access, and submit the ticket intake form.	ServiceNow Service Portal Framework
2	Transactional Intake	Renders the structured data entry form and safely translates catalog variables into target table records upon submission.	ServiceNow Record Producer
3	Dynamic Client Logic	Enforces runtime UI behavior, field visibility, and mandatory validations (e.g., Short Description constraints) on the client side.	ServiceNow UI Policies & Client Scripts

4	Data Binding & Rules	Controls the structural Category-Subcategory dependency matrix and handles direct variable-to-field data propagation.	ServiceNow Choice Lists / Variable Mapping Engine
5	Primary Database	Securely stores transactional employee issue records, schema definitions, and audit trails.	ServiceNow Relational Database Layer (`u_employee_issue`)
6	Secondary Cloud DB	Not applicable. Data persistence relies exclusively on the native ServiceNow multi-tenant database infrastructure.	Native Platform Data Layer (Implicit)
7	Enterprise File Storage	Managed by the core platform; any attachments or uploaded files are handled via native platform tables.	ServiceNow Platform System Storage (`sys_attachment`)
8	Inbound Integrations	No inbound external API connections are utilized for the core issue-creation lifecycle.	Not Applicable (Native Execution)

9	Outbound Integrations	No outbound third-party web services or API endpoints are invoked during the submission process.	Not Applicable (Native Execution)
10	Predictive Intelligence / AI	No external machine learning models, predictive engines, or AI capabilities are configured for this workflow.	Not Applicable
11	Infrastructure & Hosting	Deployed, hosted, and executed entirely within the scalable cloud infrastructure managed by ServiceNow.	ServiceNow Enterprise Cloud Platform (SaaS/PaaS)

Table-2: System & Application Attributes

S.No	**Architectural Attribute**	**Technical Specification & Implementation Strategy**	**Platform Engine / Feature**

1	Open-Source Extensibility	The system avoids external open-source frameworks to remain highly maintainable, relying strictly on standard, out-of-the-box system capabilities.	ServiceNow Platform Native Configuration
2	Security & Data Privacy	Multi-layered security framework enforcing catalog item access rules and database-level transactional security definitions.	ServiceNow User Criteria & Access Control Lists (ACLs)
3	Architecture Scalability	Built on a modular, decoupled configuration pattern allowing seamless expansion of categories, custom validations, or fields without infrastructure overhead.	ServiceNow Metadata & Application Dictionary Architecture
4	High Availability (HA)	Inherits the high availability, automated clustering, and failover capabilities of the underlying	ServiceNow Enterprise Cloud Infrastructure

		platform infrastructure.	
5	Performance & Latency	Minimizes network overhead and server round-trips via lightweight, asynchronous client-side scripts and direct database mappings.	ServiceNow Service Portal Engine & Form Rendering Engine

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References & Technical Documentation

1. **ServiceNow Documentation – Record Producer Architecture**
   Overview of data intake components and metadata configurations.
   ServiceNow Docs: Record Producer
2. **ServiceNow Documentation – Record Producer Configuration**
   Technical guidelines for constructing forms and defining catalog variables.
   ServiceNow Docs: Create a Record Producer
3. **ServiceNow Documentation – Data Propagation & Redirection**
   Best practices for scripting data mapping rules and automating user post-submission redirection.
   ServiceNow Docs: Populate Record Producer Data
4. **ServiceNow Documentation – UI Policies & Client Side Validation**
   Implementation details for dynamic form behaviors, conditional formatting, and data mandatory rules.
   ServiceNow Docs: Using UI Policies
5. **ServiceNow Documentation – Platform Access Control Layer**
   Security manuals for configuring roles, rules, and contextual data access permissions.
   ServiceNow Docs: Access Control Rules (ACL)

Would you like me to now assemble all these components—the Functional Requirements, Technical Architecture, and these Tables—into a single, consolidated Software Design Document (SDD) template in markdown format? Let me know if there are any other sections you want to add!